High-dimensional Data Visualization

II.6

Martin Theus

One of the biggest challenges in data visualization is to find general representations of data that can display the multivariate structure of more than two variables. Several graphic types like mosaicplots, parallel coordinate plots, trellis displays, and the grand tour have been developed over the course of the last three decades. Each of these plots is introduced in a specific chapter of this handbook.

This chapter will concentrate on investigating the strengths and weaknesses of these plots and techniques and contrast them in the light of data analysis problems.

One very important issue is the aspect of interactivity. Except for trellis displays, all the above plots need interactive features to rise to their full power. Some, like the grand tour, are only defined by using dynamic graphics.

6.1 Introduction

It is sometimes hard to resist the problem that is captured in the phrase "if all you have is a hammer, every problem looks like a nail." This obviously also holds true for the use of graphics. A grand tour expert will most likely include a categorical variable in the high-dimensional scatterplot, whereas an expert on mosaicplots probably will try to fit a data problem as far as possible into a categorical framework.

This chapter will focus on the appropriate use of the different plots for high-dimensional data analysis problems and contrast them by emphasizing their strengths and weaknesses.

Data visualization can roughly be categorized into two applications:

1. **Exploration**
   In the exploration phase, the data analyst will use many graphics that are mostly unsuitable for presentation purposes yet may reveal very interesting and important features. The amount of interaction needed during exploration is very high. Plots must be created fast and modifications like sorting or rescaling should happen instantaneously so as not to interrupt the line of thought of the analyst.
2. **Presentation**
   Once the key findings in a data set have been explored, these findings must be presented to a broader audience interested in the data set. These graphics often cannot be interactive but must be suitable for printed reproduction. Furthermore, some of the graphics for high-dimensional data are all but trivial to read without prior training, and thus probably not well suited for presentation purposes – especially if the audience is not well trained in statistics.

Obviously, the amount of interactivity used is the major dimension to discriminate between exploratory graphics and presentation graphics.

Interactive linked highlighting, as described by Wills (2008, Chapter II.9 same volume), is one of the keys to the right use of graphics for high-dimensional data. Linking across different graphs can increase the dimensionality beyond the number of dimensions captured in a single multivariate graphic. Thus, the analyst can choose the most appropriate graphics for certain variables of the data set; linking will preserve the multivariate context.

Although much care has been taken to ensure the best reproduction quality of all graphics in this chapter, the reader may note that the printed reproduction in black and white lacks clarity for some figures. Please refer to the book's website for an electronic version that offers full quality.

Mosaic Plots 6.2

Mosaic plots (Hartigan and Kleiner, 1981; Friendly, 1994; Hofmann, 2000) are probably the multivariate plots that require the most training for a data analyst. On the other hand, mosaicplots are extremely versatile when all possible interaction and variations are employed, as described by Hofmann (2008, Chapter III.13 same volume).

This section will explore typical uses of mosaicplots in many dimensions and move on to trellis displays.

Associations in High-dimensional Data 6.2.1

Meyer et al. (2008, Chapter III.12 same volume) already introduced techniques for visualizing association structures of categorical variables using mosaicplots. Obviously, the kind of interactions we look at in high-dimensional problems is usually more complex. Although statistical theory for categorical data often assumes that all variables are of equal importance, this may not be the case with real problems. Using the right order of the variables, mosaicplots can take the different roles of the variables into account.

Example: Detergent data

For an illustration of mosaicplots and their applications, we chose to look at the 4-D problem of the detergent data set (cf. Cox and Snell, 1991). In this data set we look at the following four variables:

1. **Water softness**
   (soft, medium, hard)
2. **Temperature**
   (low, high)
3. **M-user** (person used brand M before study)
   (yes, no)
4. **Preference** (brand person prefers after test)
   (X, M)

The major interest of the study is to find out whether or not preference for a detergent is influenced by the brand someone uses. Looking at the interaction of *M-user* and *Preference* will tell us that there is an interaction, but unrelated to the other two variables. Looking at the variables *Water Softness* and *Temperature* we will find something that is to be expected: harder water needs warmer temperatures for the same washing result and a fixed amount of detergent.

Mosaic plots allow the inspection of the interaction of *M-user* and *Preference* conditioned for each combination of *Water Softness* and *Temperature*, resulting in a plot that includes the variables in the order in which they are listed above. Figure 6.1 shows the increasing interaction of *M-user* and *Preference* for harder water and higher temperatures.

Several recommendations can be given for the construction of high-dimensional classical mosaicplots:

- The first two and the last two variables in a mosaicplot can be investigated most efficiently regarding their association. Thus the interaction of interest should be put into the last two positions of the plot. Variables that condition an effect should be the first in the plot.
- To avoid unnecessary clutter in a mosaicplot of equally important variables, put variables with only a few categories first.
- If combinations of cells are empty (this is quite common for high-dimensional data due to the curse of dimensionality), seek variables that create empty cells at high levels in the plot to reduce the number of cells to be plotted (empty cells at a higher level are not divided any further, thus gathering many potential cells into one).
- If the last variable in the plot is a binary factor, one can reduce the number of cells by linking the last variable via highlighting. This is the usual way to handle categorical response models.

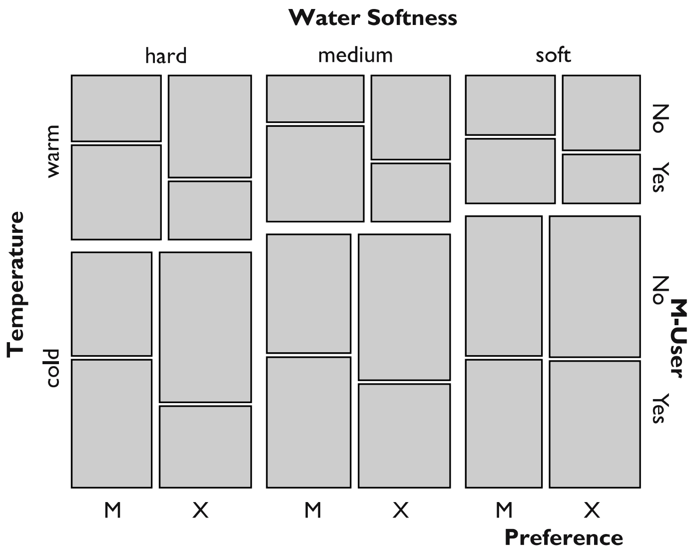


Figure 6.1. The interaction of *M-user* and *Preference* increases for harder water and higher temperatures

- Subsets of variables may reveal features far more clearly than using all variables at once. In interactive mosaicplots one can add/drop or change variables displayed in a plot. This is very efficient when looking for potential interactions between variables.

Response Models 6.2.2

In many data sets there is a single dependent categorical outcome and several categorical influencing factors. The best graphical representation of such a situation is to put all influencing factors in a mosaicplot and link the dependent variable with a barchart. This setup is shown in Fig. 6.2 for the Caesarean data set (cf. Fahrmeir and Tutz, 1994). The data set consists of three influencing factors – *Antibiotics, Risk Factor*, and *Planned* and one dependent categorical outcome, *Infection*, for 251 caesarean births. The question of interest is to find out which factors, or combination of factors, have a higher probability of leading to an infection.

At this point it is important to rethink what the highlighted areas in the mosaicplot actually show us. Let us look at the cases were no caesarean was planned, a risk factor was present, and no antibiotics were administered (the lower left cell in Fig. 6.2, which is highlighted to a high degree). In this combination, 23 of the 26 cases got an infection, making almost 88.5 %. That is

$$P(\text{Infection} \mid \overline{\text{Antibiotics}} \wedge \text{Risk Factor} \wedge \overline{\text{Planned}}) = 0.8846 .$$

But there is more we can learn from the plot. The infection probability is highest for cases with risk factors and no antibiotics administered. There is also one oddity

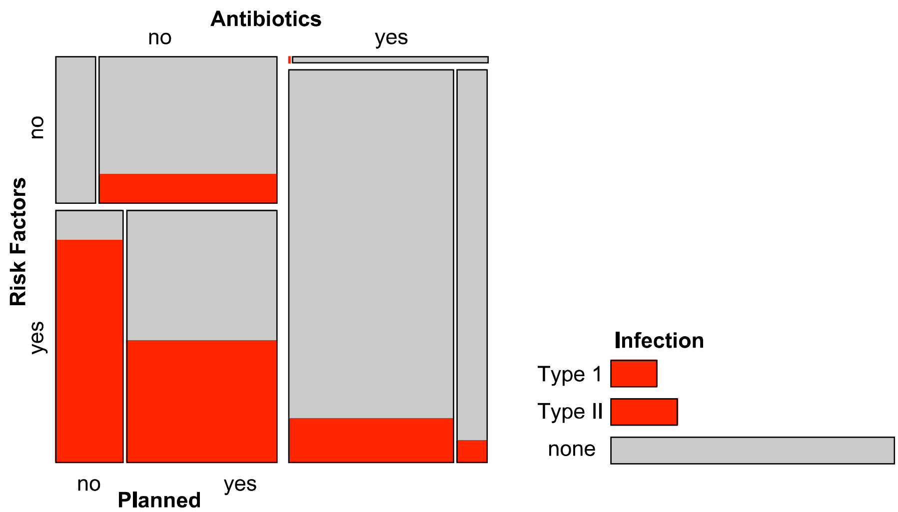


Figure 6.2. The categorical response model for the *caesarean birth* data is visualized using a mosaicplot for the influencing factors and a *barchart* for the response variable. Infection cases have been highlighted

in the data. Whereas the fact of a planned caesarean reduces the infection risk by around half, we do not have a single infection case for unplanned caesareans without risk factors and antibiotics – although at least three would be expected. Note that the chosen order is crucial for seeing this feature most easily. All these results can be investigated by looking at Fig. 6.2 but are harder to find by using classical models – nonetheless, they should be used to check significance.

6.2.3 Models

Meyer et al. (2008, Chapter III.12 same volume) presents a method for displaying association models in mosaicplots. One alternative to looking at log-linear models with mosaicplots is to plot the expected values instead of the observed values. This also permits the plotting of information for empty cells, which are invisible in the raw data but do exist in the modeled data. In general, a mosaicplot can visualize any continuous variable for crossings of categorical data, be it counts, expected values of a model, or any other positive value. Figure 6.3 shows the data from Fig. 6.1 with the two interactions *Water Softness* and *Temperature* and *M-user* and *Preference* included. Remaining residuals are coded in red (negative) and blue (positive). The feature we easily found in Fig. 6.1 – an increasing interaction between *M-user* and *Preference* – would call for a four-way interaction or at least some nonhierarchical model. Neither model can be interpreted easily or communicated to nonstatisticians. Furthermore, the log-linear model including the two two-way interactions has a p-value far greater than 0.05, suggesting that the model captures all "significant" structure. A more de-

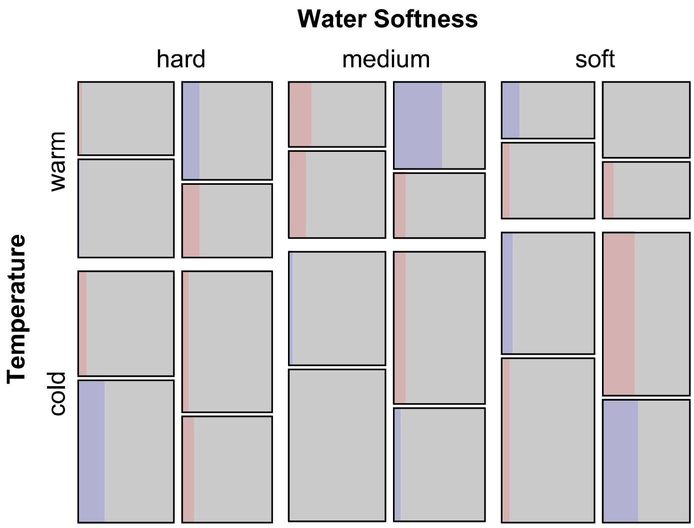


Figure 6.3. A model of the Detergent data with the interactions of *Water Softness* and *Temperature* and *M-user* and *Preference* included. The residuals are highlighted in *red* (*darker shade*) and *blue* (*lighter shade*). A very faded color indicates a high p-value

tailed discussion on log-linear models and mosaicplots can be found in Theus and Lauer (1999).

Trellis Displays

6.3

Trellis displays (called *Lattice Graphics* within the R package) also use conditioning to plot high-dimensional data. But whereas mosaicplots use a recursive layout, trellis displays use a gridlike structure to plot the data conditioned on certain subgroups.

Definition

6.3.1

Trellis displays were introduced by Becker et al. (1996) as a means to visualize multivariate data (see also Theus, 1999). Trellis displays use a latticelike arrangement to place plots onto so-called panels. Each plot in a trellis display is conditioned upon at least one other variable. To make plots comparable across rows and columns, the same scales are used in all the panel plots.

The simplest example of a trellis display is probably a boxplot y by x. Figure 6.4 shows a boxplot of the gas mileage of cars conditioned on the type of car. Results can easily be compared between car types since the scale does not change when visually traversing the different categories. Even a further binary variable can be introduced when highlighting is used, which would be the most effective way to add a third (binary) variable to the plot.

In principle, a single trellis display can hold up to seven variables at a time. Naturally five out of the seven variables need to be categorical, and two can be continuous. At the core of a trellis display we find the *panel plot*. The up to two variables plotted in the panel plot are called *axis variables*. (The current *Lattice Graphics* implementation in R does actually offer higher-dimensional plots like parallel coordinate

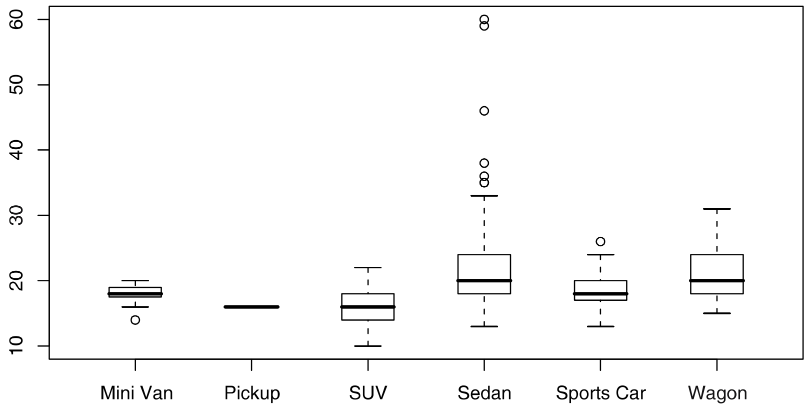


Figure 6.4. A boxplot y by x is a simple form of a trellis display

plots as panel plots, which is only a technical detail and not relevant for data analysis purposes). In principle the panel plot can be any arbitrary statistical graphic, but usually nothing more complex than a scatterplot is chosen. All panel plots share the same scale. Up to three categorical variables can be used as *conditioning variables* to form rows, columns, and pages of the trellis display. To annotate the conditioning categories of each panel plot, the so-called *strip labels* are plotted atop each panel plot, listing the corresponding category names. The two remaining variables – the so-called *adjunct variables* – can be coded using different glyphs and colors (if the panel plot is a glyph-based plot).

Trellis displays introduce the concept of *shingles*. Shingling is the process of dividing a continuous variable into (possibly overlapping) intervals in order to convert this continuous variable into a discrete variable. Shingling is quite different to conditioning with categorical variables. Overlapping shingles/intervals leads to multiple representations of data within a trellis display, which is not the case for categorical variables. Furthermore, it is hard to judge which intervals/cases have been chosen to build a shingle. Trellis displays show the interval of a shingle using an interval of the strip label. This is a solution which does not waste plotting space, but the information on the intervals is hard to read from the strip label. Nonetheless, there is a valid motivation for shingling, which is illustrated in Sect. 6.3.3.

In Fig. 6.4 we find one conditioning variable (*Car Type*) and one axis variable (*Gas Mileage*). The panel plot is a boxplot. Strip labels have been omitted as the categories can be annotated traditionally.

An example of a more complex trellis display can be found in Fig. 6.5. For the same cars data set as in Fig. 6.4, the scatterplot of *MPG* vs. *Weight* is plotted. Thus the panel plot is a scatterplot. The axis variables are *MPG* and *Weight*. The grid is set up by the two conditioning variables *Car Type* along x and *Drive* along y. A fifth variable is included as adjunct variable. The *Number of Cylinders* is included by coloring the points of the scatterplots. The upper strip label shows the category of *Drive*, the lower strip label that of *Car Type*. In Fig. 6.5 we find a common problem of trellis displays. Although the data set has almost 400 observations, 3 of the 18 panels are empty, and 3 panels have fewer than 4 observations.

6.3.2 Trellis Display vs. Mosaic Plots

Trellis displays and mosaicplots do not have very much in common. This can be seen when comparing Figs. 6.1 and 6.6. Obviously the panel plot is not a 2-D mosaicplot, which makes the comparison a bit difficult. On the other hand, the current implementations of trellis displays in R do not offer mosaicplots as panel plots, either.

In Fig. 6.6 the interaction structure is far harder to perceive than in the original mosaicplot. In a mosaicplot the presence of independence can be seen by a straight crossing of the dividing gaps of the categories (in Fig. 6.1 the user preference and the prior usage of product M can be regarded as independent for soft water and low temperatures; see lower right panel in the figure). But what does independence look like in the conditioned barchart representation of the trellis display of Fig. 6.6? Two variables in the panel plot are independent iff the ratios of all corresponding pairs of

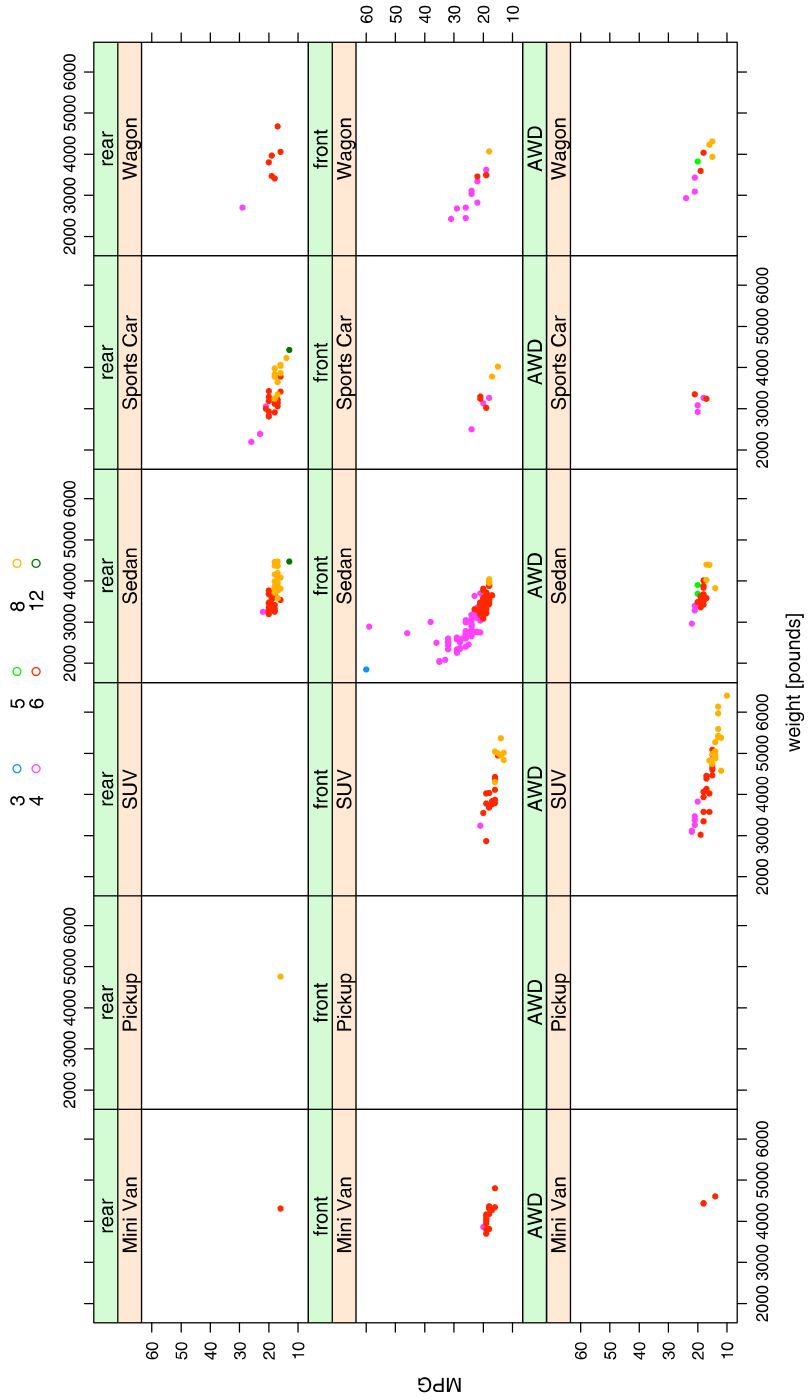


Figure 6.5. [This figure also appears in the color insert.] A trellis display incorporating five variables of the cars data set

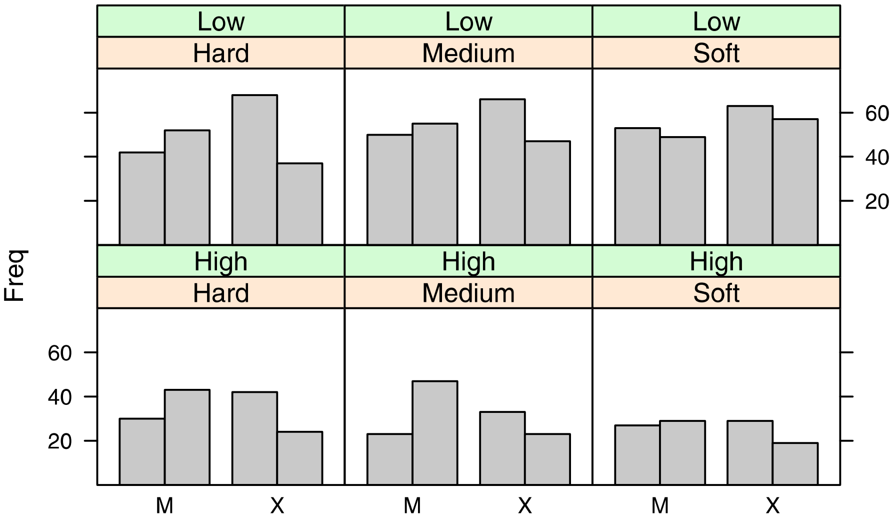


Figure 6.6. A trellis display of the detergent data from Figs. 6.1 and 6.3

levels of two variables are equal, i.e., the two barcharts are identical except for a scaling factor. Thus the only independence can be found in the panel for low temperature and soft water. Obviously it is hard to compare the ratios for more than just two levels per variable and for large absolute differences in the cell counts. Furthermore, it is even harder to quantify and compare interactions. This is because it is nontrivial to simultaneously judge the influence of differences in ratio and absolute cell sizes.

Nonetheless, there exist variations of mosaicplots (see Hofmann, 2008, Chapter III.13 same volume) that use an equal-sized grid to plot the data. Mosaic-plot vari-

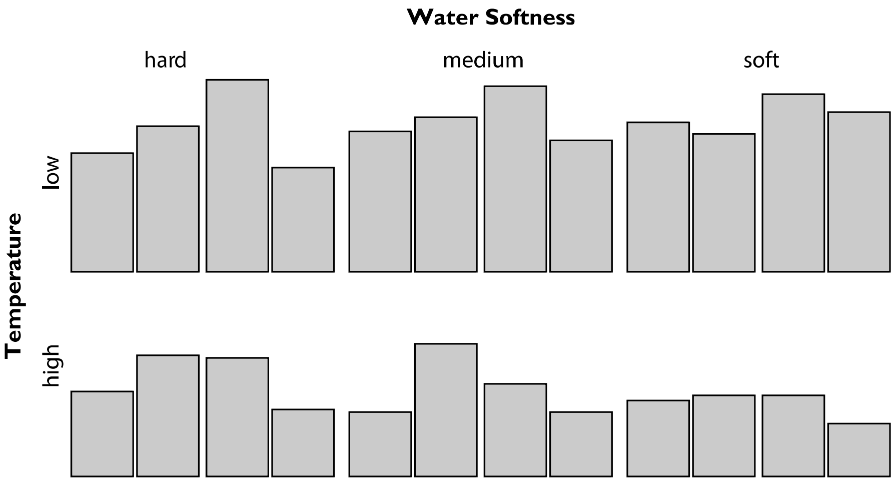


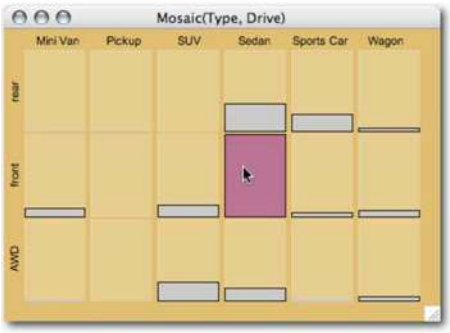
Figure 6.7. A mosaicplot in multiple multiple-barchart variation of the detergent data set that conforms exactly to the representation used in the trellis display of Fig. 6.6

ations using an equal-sized grid to plot data are *same bin size*, *fluctuation diagrams*, and *multiple barcharts*.

Figure 6.7 shows a mosaicplot in multiple multiple-barchart view with splitting directions x, y, x, x. The way the information is plotted is exactly the same as in Figs. 6.6 and 6.7. Flexible implementations of mosaicplots offering these variations can be found in Mondrian (Theus, 2002) and MANET (Unwin et al., 1996).

Trellis Displays and Interactivity 6.3.3

The conditional framework in a trellis display can be regarded as static snapshots of interactive statistical graphics. The single view in a panel of a trellis display can also be thought of as the highlighted part of the graphics of the panel plot for the conditioned subgroup. This can be best illustrated by looking at the cars data set again. Figure 6.8 shows a screenshot of an interactive session. Selecting a specific subgroup in a barchart or mosaicplot is one interaction. Another interaction would be brushing. Brushing a plot means to steadily move a brush, i.e., an indicator for the selection region, along one or two axes of a plot. The selected interval from the brush can be seen as an interval of a shingle variable. When a continuous variable is subdivided into, e.g., five intervals, this corresponds to five snapshots of the continuous brushing process from the minimum to the maximum of that variable. For the same scatterplot shown in Fig. 6.8, Fig. 6.9 shows a snapshot of a brush selecting the lowest values of the conditioning variables *Engine Size* and *Horsepower*. Now the motivation of shingle variables is more obvious, as they relate directly to this interactive technique. Brushing with linked highlighting is certainly far more flexible than the static view in a trellis display. On the other hand, the trellis display can easily be reproduced in printed form, which is impossible for the interactive process of brushing.



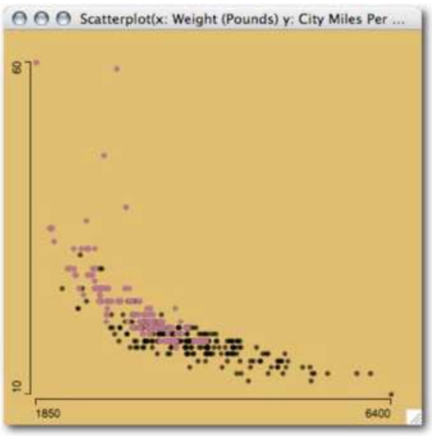


Figure 6.8. Selecting the group of front-wheel-drive sedans in the mosaicplot in multiple-barchart view (*left*), one gets the corresponding panel plot (scatterplot on *right*) from Fig. 6.5 in the highlighted subgroup

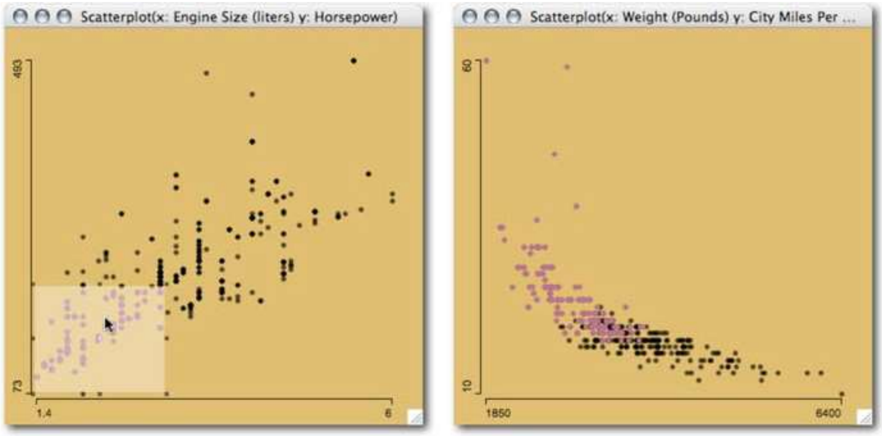


Figure 6.9. Brushing in the conditioning scatterplot (*left*), one gets the panel plot (scatterplot on *right*) from Fig. 6.5 in the highlighted subgroup

6.3.4 Visualization of Models

The biggest advantage of trellis displays is the common scale among all plot panels. This allows an effective comparison of the panel plots between rows, columns, and pages, depending on the number of conditioning variables and the type of panel plot. Trellis displays are most powerful when used for model diagnostics. In model diagnostics one is most interested in understanding for what data the model fits well and for which cases it does not.

In a trellis display the panel plot can incorporate model information like fitted curves or confidence intervals conditioned for exactly the subgroup shown in the panels. For each panel, the fit and its quality can then be investigated along with the raw data. Figure 6.10 shows the same plot as in Fig. 6.5 except for the adjunct variable. Each scatterplot has a lowess smoother superimposed. One problem with trellis displays is the fact that it is hard to judge the number of cases in a panel plot. For example, in Fig. 6.10 it would be desirable to have confidence bands for the scatterplot smoother in order to be able to judge the variability of the estimate across panels.

Wrap-up

As can be seen from the examples in this section, trellis displays are most useful for continuous axis variables, categorical conditioning variables, and categorical adjunct variables. Shingling might be appropriate under certain circumstances, but it should generally be avoided to ease interpretability.

The major advantage of trellis displays over other multivariate visualization techniques is the flat learning curve of such a display and the possibilities of static reproduction as current trellis display implementations do not offer any interactions. Trellis displays also offer the possibility of easily adding model information to the plots.

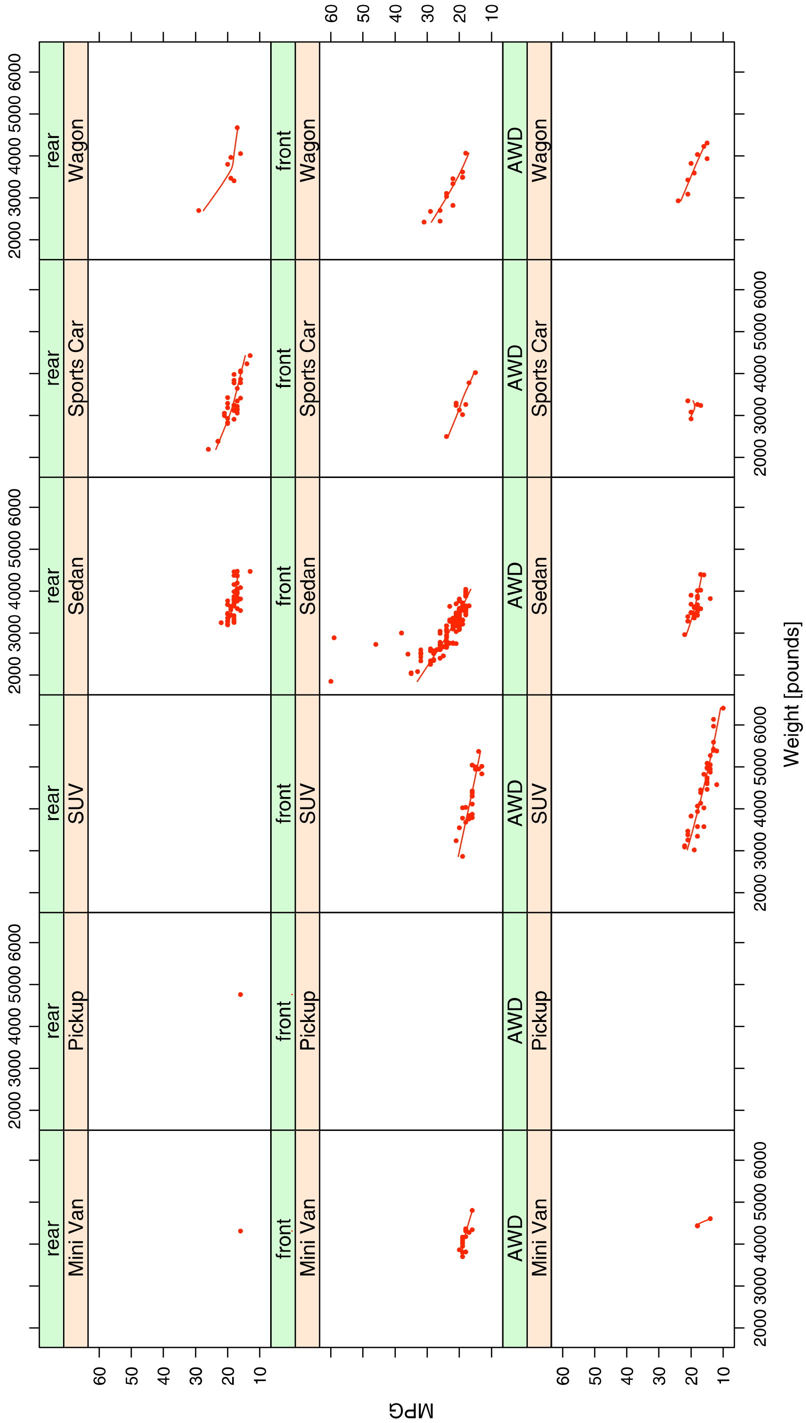


Figure 6.10. The same trellis display as in Fig. 6.5 with an additional lowess smoother superimposed

Nevertheless, interactive linked graphics are usually more flexible in exploratory data analysis applications. Linking the panel plot to barcharts or mosaicplots of the conditioning variables and/or adjunct variables or brushing over a shingle variable is more flexible, though these techniques lack the global overview and the possibility of static reproduction.

6.4 Parallel Coordinate Plots

Parallel coordinate plots, as described by Inselberg (2008, Chapter III.14 same volume), escape the dimensionality of two or three dimensions and can accommodate many variables at a time by plotting the coordinate axes in parallel. They were introduced by Inselberg (1985) and discussed in the context of data analysis by Wegman (1990).

6.4.1 Geometrical Aspects vs. Data Analysis Aspects

Whereas in Inselberg (2008, Chapter III.14 same volume), the geometrical properties of parallel coordinate plots are emphasized to visualize properties of high-dimensional data-mining and classification methods, this section will investigate the main use of parallel coordinate plots in data analysis applications. The most interesting aspects in using parallel coordinate plots are the investigation of groups/clusters, outliers, and structures over many variables at a time. Three main uses of parallel coordinate plots in exploratory data analysis can be identified as the following:

- **Overview**
  No other statistical graphic can plot so much information (cases and variables) at a time. Thus parallel coordinate plots are an ideal tool to get a first overview of a data set. Figure 6.11 shows a parallel coordinate plot of almost 400 cars with 10 variables. All axes have been scaled to *min-max*. Several features, like a few very expensive cars, three very fuel-efficient cars, and the negative correlation between car size and gas mileage, are immediately apparent.
- **Profiles**
  Despite the overview functionality, parallel coordinate plots can be used to visualize the profile of a single case via highlighting. Profiles are not only restricted to single cases but can be plotted for a whole group, to compare the profile of that group with the rest of the data.
  Using parallel coordinate plots to profile cases is especially efficient when the coordinate axes have an order like time.
  Figure 6.12 shows an example of a single profile highlighted – in this case, the most fuel-efficient car.

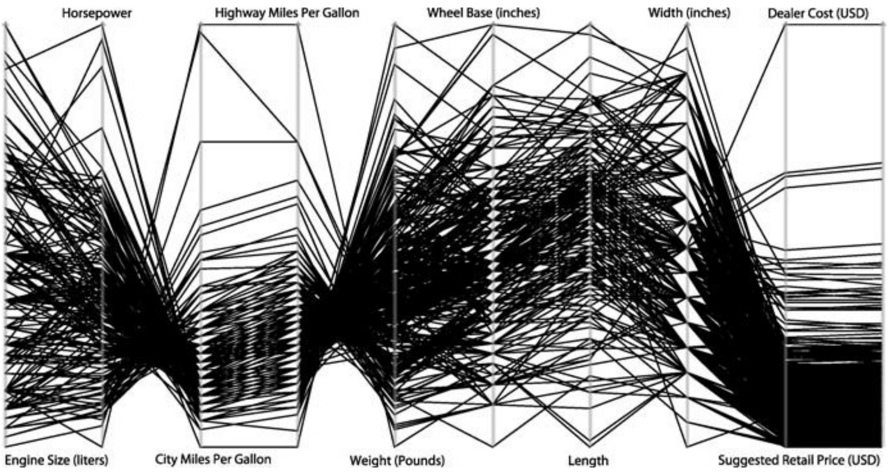


Figure 6.11. Parallel coordinate plot for 10 variables on almost 400 cars

– **Monitor**
When working on subsets of a data set parallel coordinate plots can help to relate features of a specific subset to the rest of the data set. For instance, when looking at the result of a multidimensional scaling procedure, parallel coordinate plots can help to find the major axes, which influence the configuration of the MDS. Figure 6.13 shows a 2-D MDS along with the corresponding parallel coordinate plot. Querying the leftmost cases in the MDS shows that these cars are all hybrid cars with very high gas mileage. The top right cases in the MDS correspond to heavy cars like pickups and SUVs. Obviously, similar results could have been found with biplots.

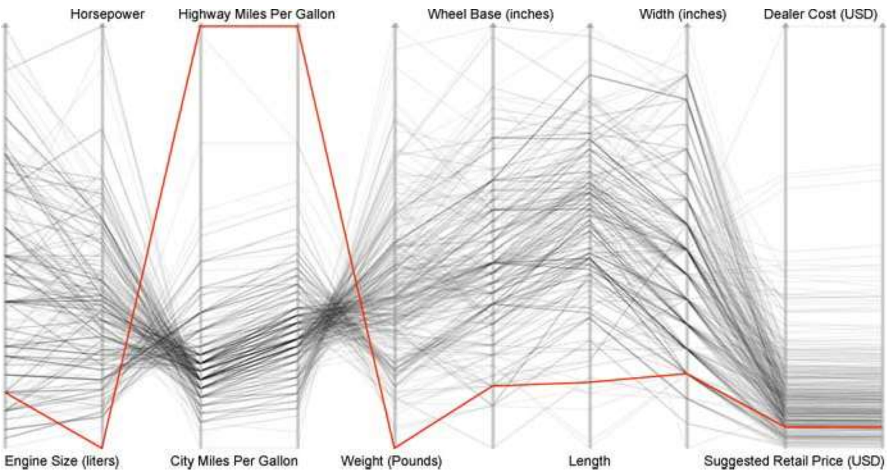


Figure 6.12. PCP from Fig. 6.11 with the most fuel-efficient car (Honda Insight 2dr.) highlighted

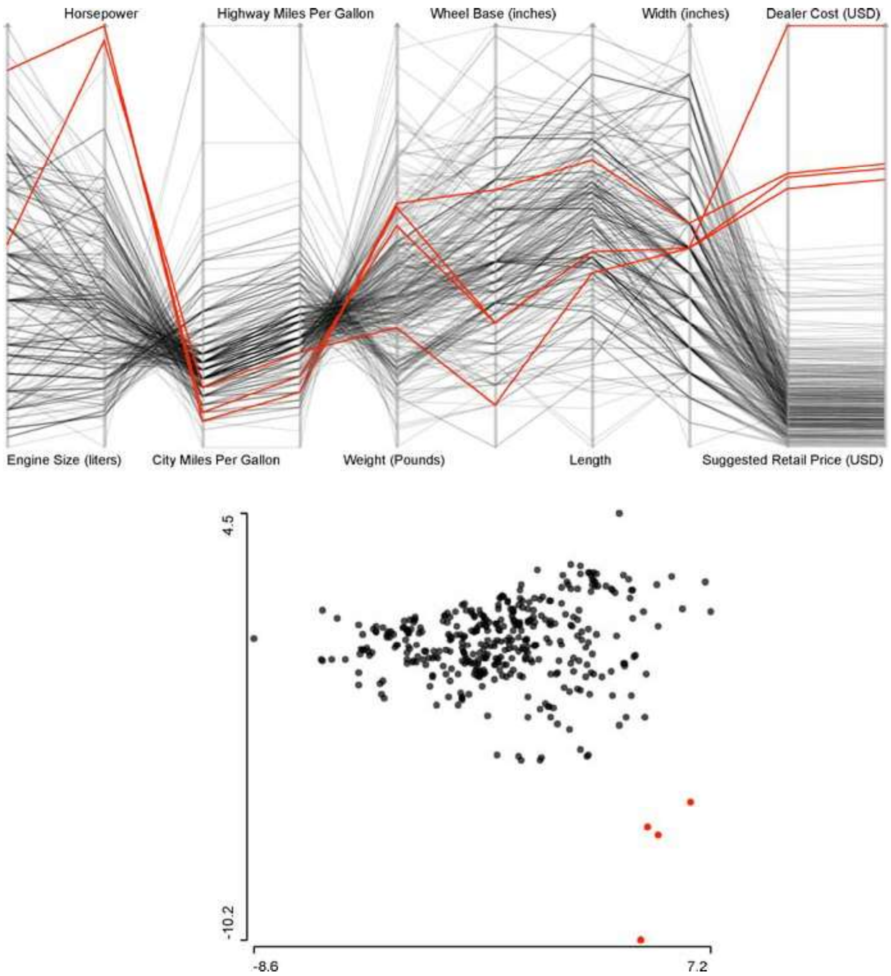


Figure 6.13. MDS (*bottom*) and parallel coordinate plot (*top*) of the cars data set. The four cars in the *lower right* of the MDS are highlighted and happen to be the most expensive and most powerful cars in the data set

6.4.2 Limits

Parallel coordinates are often overrated with respect to the insight they provide into multivariate features of a data set. Obviously scatterplots are superior for investigating 2-D features, but scatterplot matrices (SPLOMs) need far more space to plot the same information as PCPs. Even the detection of multivariate outliers is not something that can usually be directly aided by parallel coordinates. Detecting features in parallel coordinates that are **not** visible in a 1-D or 2-D plot is rare. On the other hand, parallel coordinate plots are extremely useful for interpreting the findings of multivariate procedures like outlier detection, clustering, or classification in a highly multivariate context.

Although parallel coordinate plots can handle many variables at a time, their rendering limits are reached very soon when plotting more than only a few hundreds of lines. This is due to overplotting, which is far worse than with scatterplots, since parallel coordinate plots only use one dimension to plot the information and the glyph used (a line) prints far more ink than the glyphs in a scatterplot (points). One solution to coping with overplotting is to use α-blending. When α-blending is used, each polygon is plotted with only α% opacity, i.e., $(1-\alpha)$% transparency. With smaller α values, areas of high line density are more visible and hence are better contrasted to areas with a small density.

Figures 6.11 to 6.13 use α-blending to make the plots better readable or to emphasize the highlighted cases. In the cars data set we only look at fewer than 400 cases, and one can imagine how severe the overplotting will get once thousands of polylines are plotted.

Figures 6.14 and 6.15 show two examples of how useful α-blending can be. The so-called "Pollen" data used in Fig. 6.14 come from an ASA data competition in the late 80s. The data are completely artificial and have the word "E U R E K A" woven into the center of the simulated normal distributions. The almost 4000 cases in five dimensions produce a solid black band without any α-blending applied. Going down to an alpha value of as little as 0.005 will reveal a more solid thin line in the center of the data. Zooming in on these cases will find the word "Eureka," which just increases the simulated density in the center enough to be visible.

The data in Fig. 6.15 are real data from Forina et al. (1983) on the fatty acid content of Italian olive oil samples from nine regions. The three graphics show the same plot of all eight fatty acids with α-values of 0.5, 0.1, and 0.05. Depending on the amount of α-blending applied, the group structure of some of the nine regions is more or less visible.

Note that it is hard to give general advice on how much α-blending should be applied because the rendering system and the actual size of the plot may change its

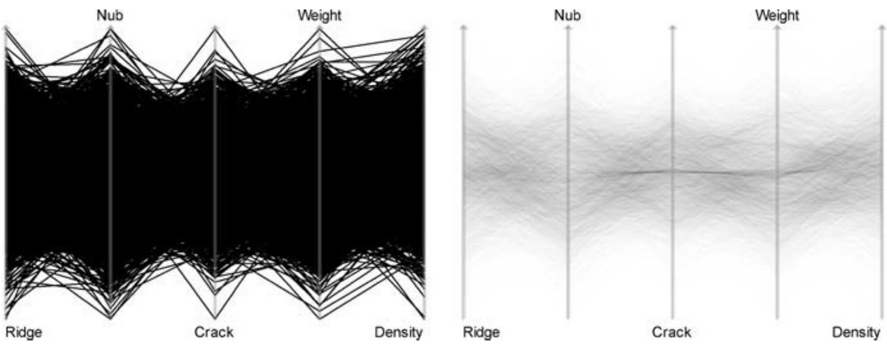


Figure 6.14. The "Pollen" data with $\alpha = 1$ (*left*) and $\alpha = 0.005$ (*right*)

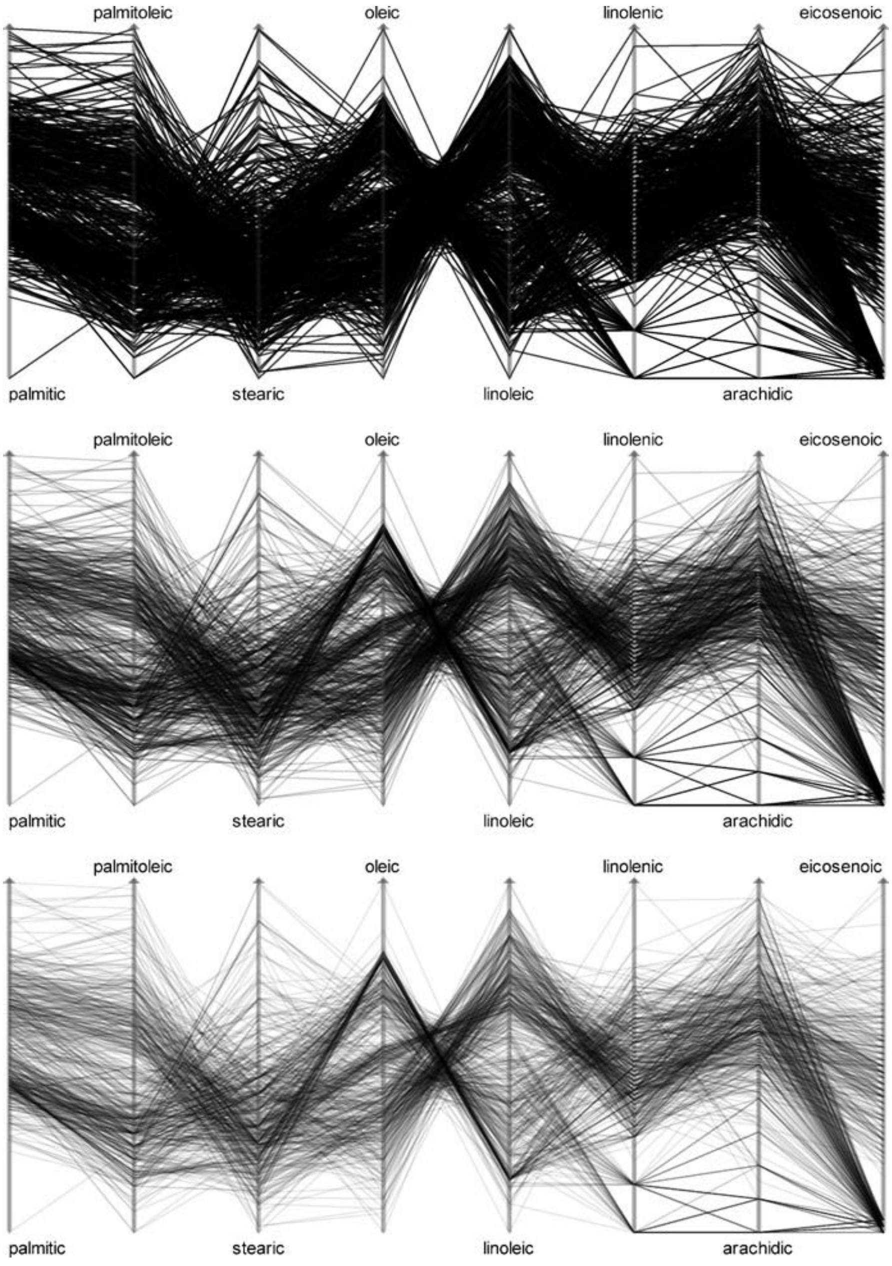


Figure 6.15. The "Olive Oils" data with $\alpha = 0.5$ (*top*), $\alpha = 0.1$ (*middle*), and $\alpha = 0.05$ (*bottom*)

appearance substantially. As with many exploratory techniques, the user should experiment with different settings until he or she feels comfortable enough with the insight gained.

Sorting and Scaling Issues 6.4.3

Parallel coordinate plots are especially useful for variables which either have an order such as time or all share a common scale. In these cases, scaling and sorting issues are very important for a successful exploration of the data set.

Sorting

Sorting in parallel coordinate plots is crucial for the interpretation of the plots, as interesting patterns are usually revealed at neighboring variables. In a parallel coordinate plot of k variables, only $k-1$ adjacencies can be investigated without reordering the plot. The default order of a parallel coordinate plot is usually the sequence in which the variables are passed to the plotting routine, in most cases the sequence in the data file itself. In many situations this order is more or less arbitrary. Fortunately, one only needs $\lfloor \frac{k+1}{2} \rfloor$ different orderings to see all adjacencies of k variables (see Wegman, 1990).

Whenever all, or at least groups of, variables share the same scale, it is even more helpful to be able to sort these variables according to some criterion. This can be statistics of the variables (either all cases or just a selected subgroup) like minimum, mean, range, or standard deviation, the result of a multivariate procedure, or even some external information. Sorting axes can reduce the visual clutter of a parallel coordinate plot substantially.

If data sets are not small, sorting options have to be provided both manually and automatically.

Scalings

Besides the default scaling, which is to plot all values over the full range of each axis between the minimum and the maximum of the variable, several other scalings are useful. The most important scaling option is to either individually scale the axes or to use a common scale over all axes. Other scaling options define the alignment of the values, which can be aligned at:

- The mean
- The median
- A specific case
- A specific value

For an aligned display, it is not obvious what the range of the data should be when an individual scale is chosen. For individual scales, a 3σ scaling is usually a good choice to map the data onto the plot area.

Alignments do not force a common scale for the variables. Common scaling and alignments are independent scaling options.

Figure 6.16 shows a parallel coordinate plot for the individual stage times of the 155 cyclists who finished the 2005 Tour de France bicycle race. In the upper plot we

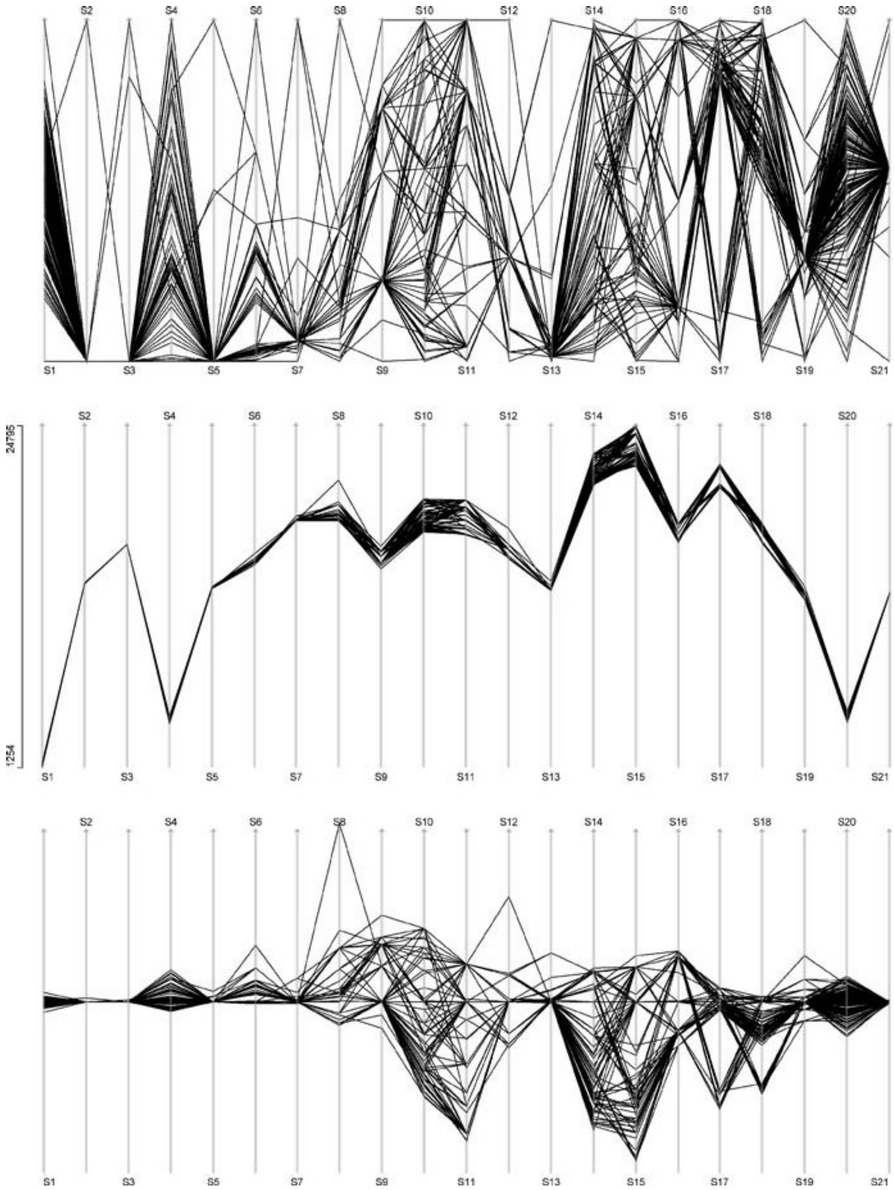


Figure 6.16. Three scaling options for the stage times in the Tour de France 2005: *Top*: all stages are scaled individually between minimum and maximum value of the stage (usual default for parallel coordinate plots) *Middle*: a common scale is used, i.e., the minimum/maximum time of all stages is used as the global minimum/maximum for all axes (this is the only scaling option where a global and common axis can be plotted) *Below*: common scale for all stages, but each stage is aligned at the median value of that stage, i.e., differences are comparable, locations not

see the default min-max scaling. Except for some local groups, not much can be seen from this plot. The middle plot shows the common scaling option for the same data. Now the times are comparable, but due to the differences in absolute time needed for a short time trial and a hard mountain stage, the spread between the first and the last cyclist is almost invisible for most of the stages. Again, except for some outliers, there is hardly anything to see in this representation. The lower plot in Fig. 6.16 shows the same data as the upper two plots, but now each axis is aligned at its median (the median has the nice interpretation of capturing the time of the peloton). Note that the axes still have the same scale, i.e., time differences are still comparable, but now are aligned at the individual medians. This display option clearly reveals the most information.

For a better description of the race as a whole, it is sensible to look at the cumulative times instead of the stage times. Figure 6.17, left, shows a parallel coordinate plot for the cumulative times for each of the 155 cyclists who completed the tour for the corresponding stage. The scaling is the typical default scale usually found in parallel coordinate plots, i.e., individually scaled between minimum and maximum of each axis. All drivers of the team "Discovery Channel" are selected. Although this scaling option gives the highest resolution for these data, it is desirable to have a common scale for all axes. A simple common scale won't do the trick here, as the cumulative times keep growing, dwarfing the information of the early stages. Figure 6.17, right, uses a common scale, but additionally each axis is aligned at the median of each variable. (Time differences at early stages are not very interesting for the course of the race). Figure 6.17, right, now shows nicely how the field spreads from stage to stage and how the mountain stages (e.g., stages 14 to 16 are stages in the Alps) spread the field far more than flat stages. The drivers of the team "Discovery Channel" are also selected in this plot, showing how the team was separated during the course of the race, although most of the cyclists remained in good positions, supporting the later winner of the race.

The development of the race can be compared in Fig. 6.18, where the plot from Fig. 6.17, right, is shown along with two profile plots. The upper profile plot shows the cumulative category of the stage, which is the sum of 5 minus the category of a mountain in the stage. The peaks starting at stages 8 and 14 nicely indicate the mountain stages in the Pyrenees and the Alps. The lower profile plot gives the average speed of the winner of each stage. Obviously both profiles are negatively correlated.

Wrap-up 6.4.4

Parallel coordinate plots are not very useful "out of the box," i.e., without features like α-blending and scaling options. The examples used in this chapter show how valuable these additions are in order to get a sensible insight into high-dimensional continuous data. Highlighting subgroups can give additional understanding of group structures and outliers.

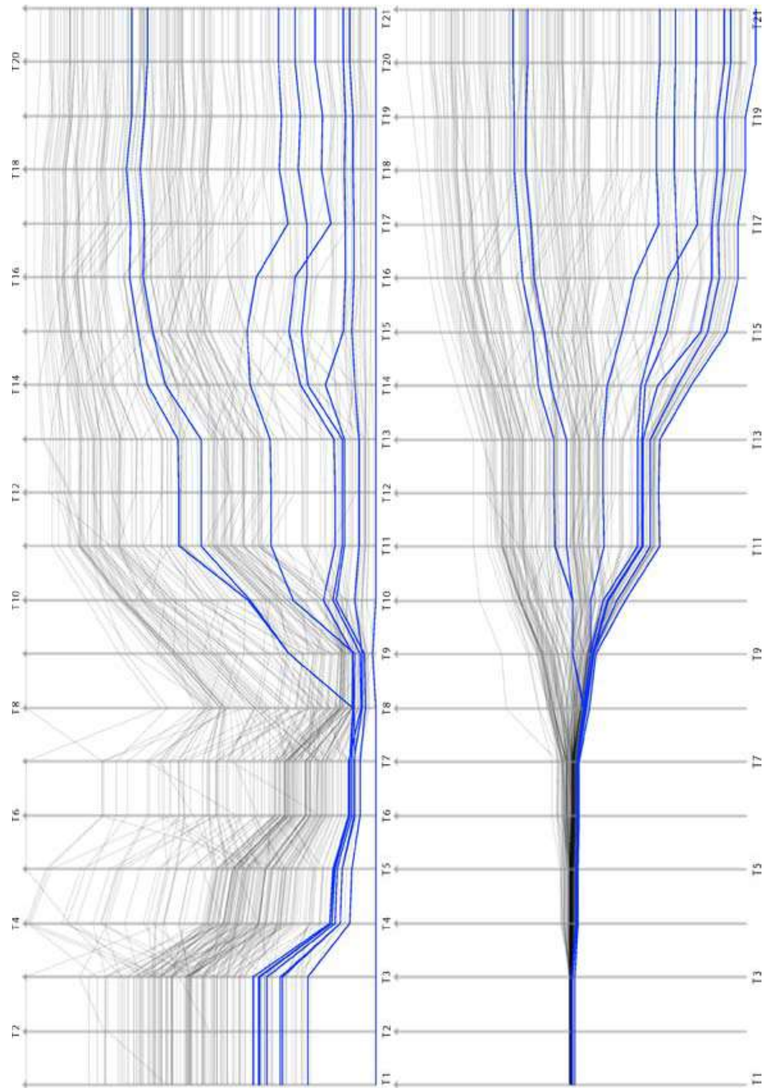


Figure 6.17. Results from 2005 Tour de France. *Left*: each *axis* shows the cumulative results for all 155 cyclists who finished the tour. *Right*: common scaling applied and *axes* aligned at medians of each stage. (The eight riders from team "Discovery Channel" are selected)

6.5 Projection Pursuit and the Grand Tour

The grand tour (see Buja and Asimov, 1986) is by definition (see Chen et al., 2008, Chapter III.2) a purely interactive technique. Its basic definition is:

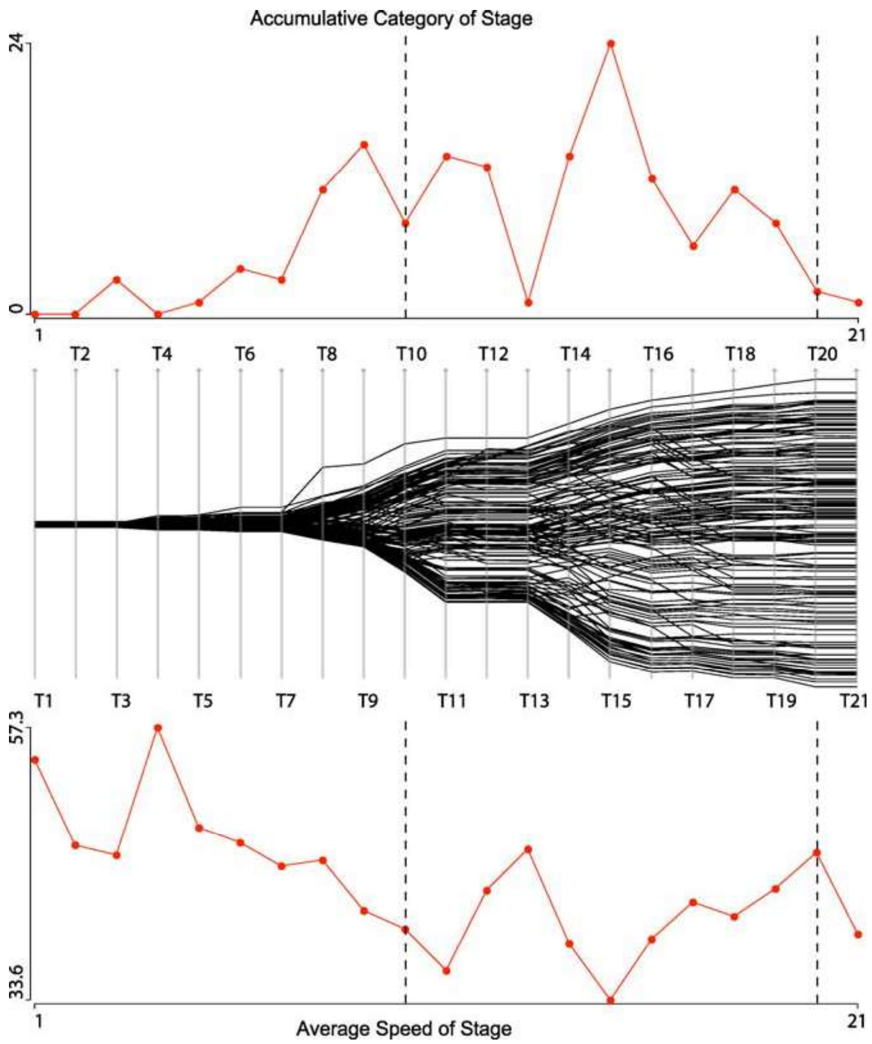


Figure 6.18. The same plot as Fig. 6.17, right, shown in the *middle*, with a profile plot of the cumulative category of the mountains in the stage (*top*) and the average speed (of the winner) of each stage (*bottom*)

> A continuous 1-parameter family of d-dimensional projections of p-dimensional data that is dense in the set of all d-dimensional projections in $\mathbb{R}^p$. The parameter is usually thought of as time.

For a 3-D rotating plot, parameter p equals 3 and parameter d equals 2. In contrast to the 3-D rotating plot, the grand tour does not have classical rotational controls but uses successive randomly selected projections. Figure 6.19 shows an example of three successive planes $P1$, $P2$, and $P3$ in three dimensions.

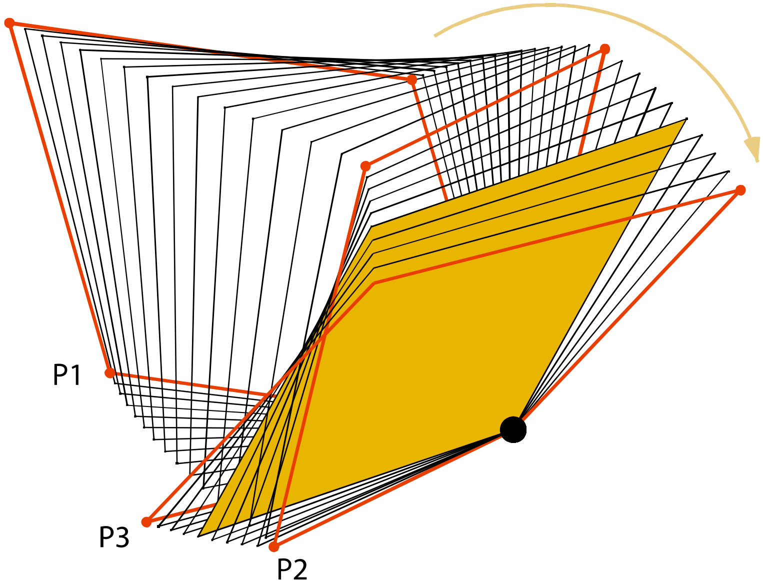


Figure 6.19. Example path of a grand tour

The planes between the randomly selected base planes are interpolated to get a smooth pseudorotation, which is comparable to a physical 3-D rotation. A more technical description of the grand tour can be found in Buja et al. (1999). Although the human eye is not very well trained to recognize rotations in more than three dimensions, the grand tour helps reveal structures like groups, gaps, and dependencies in high-dimensional data, which might be hard to find along the orthogonal projections.

Although projections are usually monitored using a scatterplot, any other plot like a histogram or parallel coordinate plot can be used to display the projected data (see, e.g., Wegman, 1991).

Projection pursuit is a means to get more guidance during the rotation process. A new projection plane is selected by optimizing a projection pursuit index, which measures a feature like point mass, holes, gaps, or other target structures in the data.

6.5.1 Grand Tour vs. Parallel Coordinate Plots

The grand tour is a highly exploratory tool, even more so than the other methods discussed in this chapter. Whereas in classical parallel coordinate plots the data are still projected to orthogonal axes, the grand tour permits one to look at arbitrary nonorthogonal projections of the data, which can reveal features invisible in orthogonal projections. Figure 6.20 shows an example of three projections of the cars data set using the grand tour inside ggobi (see Swayne et al., 2003). The cases are colored according to the number of cylinders. Each plot has the projection directions of the 11 variables added at the lower left of the plot. Obviously these static screenshots of

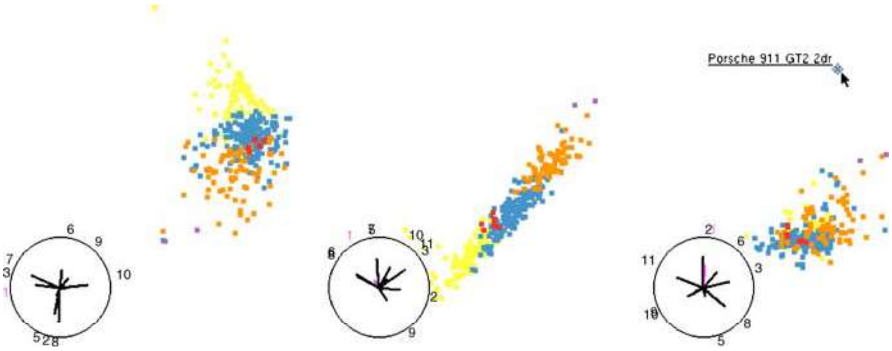


Figure 6.20. Three example screenshots of three different projections of the cars data set. The cases are colored according to the number of cylinders. The *rightmost plot* has the least discrimination of the groups but the strongest separation of an outlier, the "Porsche GT"

the projections are not very satisfactory unless they reveal a striking feature. Whereas the parallel coordinate plot of the same data (cf. Fig. 6.11) can at least show the univariate distributions along with some bivariate relationships, the grand tour fails to do so, and focuses solely on the multivariate features, which may be visible in certain projections.

The grand tour can help to identify the geometry of variables beyond the limits of the three dimensions of a simple rotating plot. Nevertheless, examples of structures in more than five dimensions are rare, even when using the grand tour. In these cases the fixed geometrical properties of parallel coordinate plots seem to be an advantage.

Monitoring the projected data in parallel coordinates instead of a simple scatterplot is a promising approach to investigating data beyond ten or even more dimensions. Unfortunately, only very few flexible implementations of the grand tour and projection pursuit exist, which limits the possibility of a successful application of these methods.

Recommendations 6.6

This chapter showed the application, strengths, and weaknesses of the most important high-dimensional plots in statistical data visualization. All plots have their specific field of application, where no other method delivers equivalent results. The fields of application are broader or narrower depending on the method. All four methods and techniques discussed in this chapter, i.e., mosaicplots, trellis displays, parallel coordinate plots, and the grand tour and projection pursuit, need a certain amount of training to be effective. Furthermore, training is required to learn the most effective use of the different methods for different tasks.

Some of the plots are optimized for presentation graphics (e.g., trellis displays), others, in contrast only make sense in a highly interactive and exploratory setting (e.g., grand tour).

The high-dimensional nature of the data problems for the discussed plots calls for interactive controls – be they rearrangements of levels and/or variables or different scalings of the variables or the classical linked highlighting – which put different visualization methods together into one framework, thus further increasing the dimensionality.

Figure 6.21 illustrates the situation. When analyzing high-dimensional data, one needs more than just one visualization technique. Depending on the scale of the variables (discrete or continuous) and the number of variables that should be visualized simultaneously, one or another technique is more powerful. All techniques – except for trellis displays – have in common that they only rise to their full power when interactive controls are provided. Selection and linking between the plots can bring the different methods together, which then gives even more insight. The results found in an exploration of the data may then be presented using static graphics. At this point, trellis displays are most useful to communicate the results in an easy way.

Finally, implementations of these visualization tools are needed in software. Right now, most of them are isolated features in a single software package. Trellis displays can only be found in the R package, or the corresponding commercial equivalent

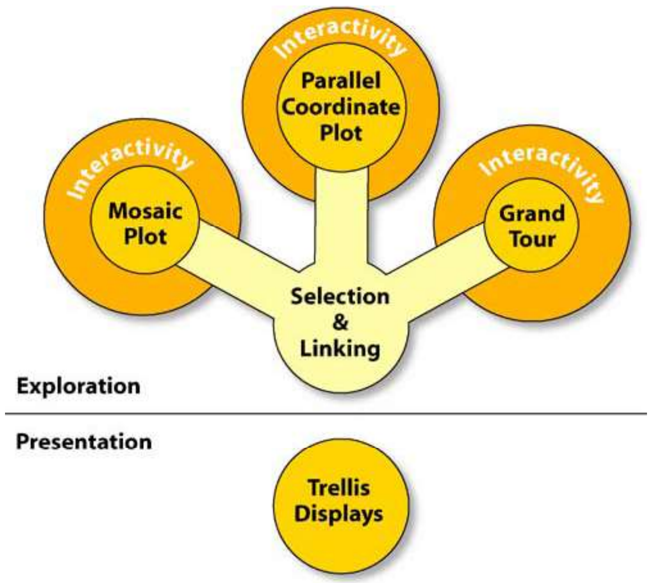


Figure 6.21. Diagram illustrating the importance of interactivity and linking of the high-dimensional visualization tools in statistical graphics

S-Plus, and grand tour and projection pursuit only in the ggobi package. It would be desirable to have a universal tool that could integrate all of the methods in a highly interactive and tightly linked way.

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